

Discover more

K-Map practice problems

Computer science textbook

maps

Karnaugh Map

Number of Variables:

4 Variables

Clear All

Fill All with 1

CD

AB

00	00	01	11	10
00	1 0	0 1	0 3	0 2
01	0 4	1 5	1 7	0 6
11	0 12	1 13	1 15	1 14
10	0 8	0 9	0 11	0 10

Click cells to toggle values: 0 → 1 → X → 0

X represents "don't care" condition (can be either 0 or 1)

Simplified K-Map Result

SOP Form

POS Form

CD

AB

00	01	11	10	
00	1 0	0 1	0 3	0 2
01	0 4	1 5	1 7	0 6
11	0 12	1 13	1 15	1 14
10	0 8	0 9	0 11	0 10

Simplified Boolean Expression

A'B'C'D' + ABC + BD

Original Groups

Original

A'B'C'D'

Minterms

(0)

Simplified

A'B'C'D'

Original

ABCD' + ABCD

Minterms

(14, 15)

Simplified

ABC

Original

A'BC'D + A'BCD + ABC'D + ABCD

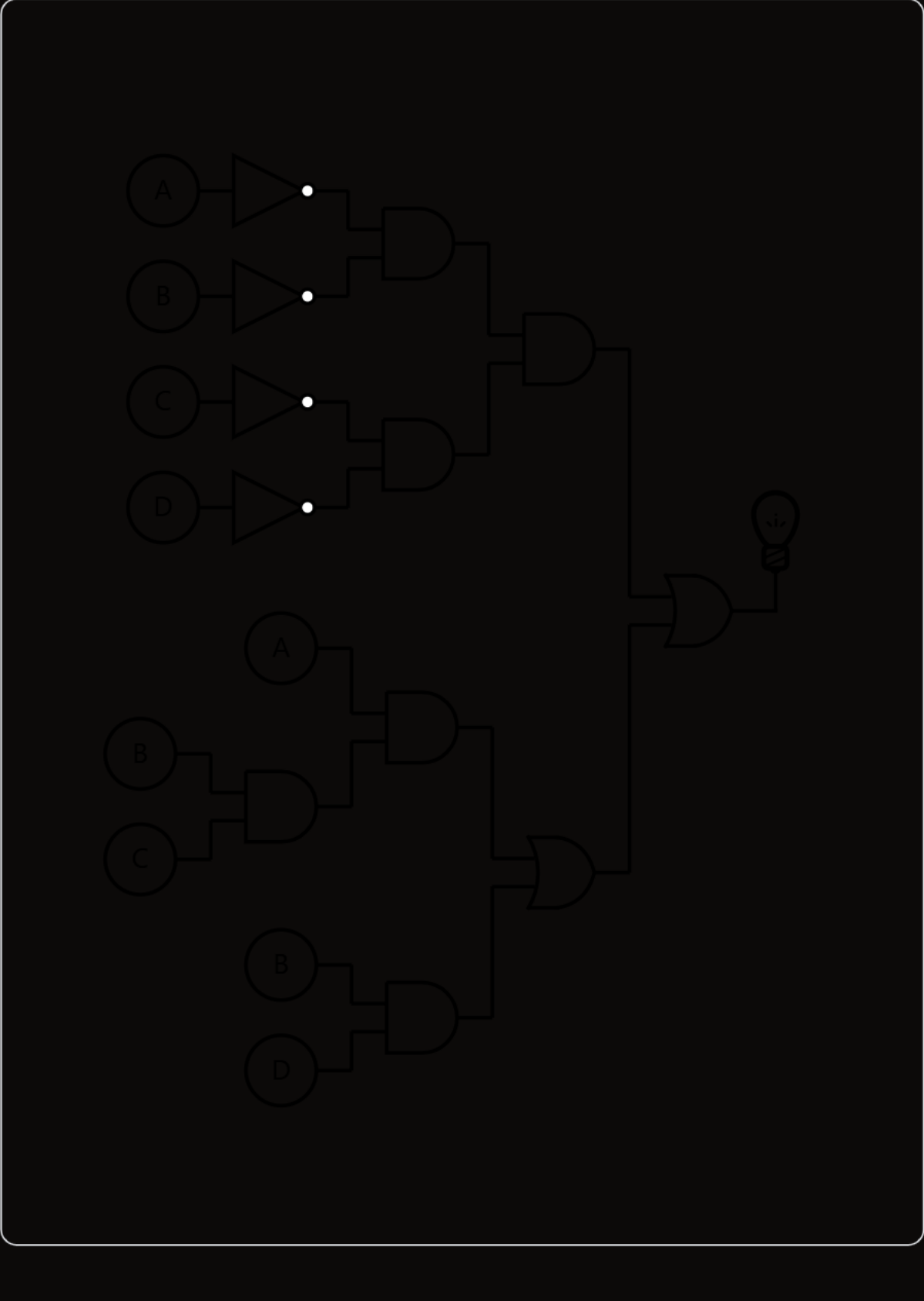
Minterms

(5, 7, 13, 15)

Simplified

BD

Logic Circuit



Truth Table

#	A	B	C	D	Output		
0	0	0	0	0	0	1	X
1	0	0	0	1	0	1	X
2	0	0	1	0	0	1	X
3	0	0	1	1	0	1	X
4	0	1	0	0	0	1	X
5	0	1	0	1	0	1	X
6	0	1	1	0	0	1	X
7	0	1	1	1	0	1	X
8	1	0	0	0	0	1	X
9	1	0	0	1	0	1	X
10	1	0	1	0	0	1	X
11	1	0	1	1	0	1	X
12	1	1	0	0	0	1	X
13	1	1	0	1	0	1	X
14	1	1	1	0	0	1	X
15	1	1	1	1	0	1	X

Click buttons to set output values: 0 or 1 or X

X represents "don't care" condition (can be either 0 or 1)

Minterms

Maxterms